

# Unit-I

## Introduction to Artificial Intelligence

### **Definition and Scope of Artificial Intelligence (AI)**

#### **Definition of Artificial Intelligence**

**Artificial Intelligence (AI)** is a branch of computer science that deals with the development of **machines and computer systems capable of performing tasks that normally require human intelligence.**

These tasks include:

- Learning from experience
- Reasoning and problem-solving
- Understanding natural language
- Recognizing images and speech
- Decision-making
- Planning and prediction

#### **Simple definition:**

AI is the ability of a machine or computer program to imitate human intelligence and perform intelligent tasks.

#### **Scope of Artificial Intelligence**

The scope of AI is very broad and covers many areas:

1. **Machine Learning (ML)**  
Enables computers to learn from data and improve their performance without being explicitly programmed.
2. **Natural Language Processing (NLP)**  
Helps computers understand, interpret, and generate human language.  
*Example: Chatbots and language translation.*
3. **Computer Vision**  
Enables machines to identify and understand objects, faces, images, and videos.
4. **Robotics**  
Uses AI to develop machines that can sense their environment, make decisions, and perform physical tasks.
5. **Expert Systems**  
AI systems that use knowledge and rules to solve problems in specific domains, such as medical diagnosis.

6. **Speech Recognition**  
Converts human speech into machine-understandable text or commands.
7. **Planning and Decision Making**  
AI can analyze different possibilities and select suitable actions to achieve a particular goal.
8. **Game Playing**  
AI is used in games such as chess and strategy games to make intelligent decisions.
9. **Healthcare**  
AI assists in disease detection, medical-image analysis, drug discovery, and personalized treatment.
10. **Finance and Business**  
AI is used for fraud detection, risk analysis, recommendation systems, and customer support.

AI aims to create **intelligent systems that can perceive, learn, reason, and act**. Its applications range from everyday technologies such as virtual assistants and recommendation systems to advanced areas such as autonomous vehicles, healthcare, robotics, and scientific research.

## Historical Context and Evolution of Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science concerned with creating machines and software capable of performing tasks that normally require human intelligence, such as **learning, reasoning, problem-solving, perception, language understanding, and decision-making**.

### Historical Context of AI

The idea of artificial beings with human-like intelligence is very old. However, AI became a scientific field only after the development of modern computers.

### Major stages in the evolution of AI

| Period      | Development                                                                    |
|-------------|--------------------------------------------------------------------------------|
| 1940s–1950s | Development of electronic computers and early ideas about machine intelligence |
| 1950        | Alan Turing proposed the <b>Turing Test</b>                                    |
| 1956        | Dartmouth Conference; the term <b>Artificial Intelligence</b> was introduced   |
| 1950s–1960s | Development of symbolic AI, logic programs and problem-solving systems         |
| 1970s–1980s | Expert systems became popular                                                  |

| Period      | Development                                                          |
|-------------|----------------------------------------------------------------------|
| 1980s–1990s | AI experienced funding reductions, known as <b>AI winters</b>        |
| 1990s–2000s | Machine learning became increasingly important                       |
| 2010s       | Deep learning achieved major advances in vision, speech and language |
| 2020s       | Generative AI and large language models became widely used           |

## Dartmouth Conference – 1956

The **Dartmouth Summer Research Project on Artificial Intelligence** in 1956 is generally considered the birth of AI as an academic field. Researchers including **John McCarthy, Marvin Minsky, Claude Shannon, and Nathaniel Rochester** proposed studying whether aspects of intelligence could be described precisely enough for machines to simulate them.

## Turing Test

The **Turing Test** was proposed by British mathematician and computer scientist **Alan Turing** in his 1950 paper *Computing Machinery and Intelligence*.

Instead of asking the difficult question “**Can machines think?**”, Turing proposed an operational test based on a machine's ability to communicate like a human.

### Basic idea

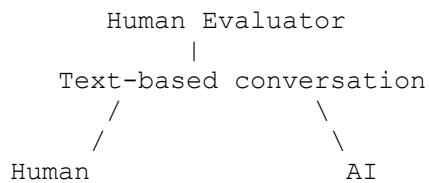
The test involves:

- A **human evaluator**
- A **human participant**
- A **machine**

The evaluator communicates with both through text, without knowing which participant is the machine.

If the evaluator cannot reliably distinguish the machine from the human based on their responses, the machine is considered to have demonstrated human-like conversational intelligence

## Simple representation



## Importance of the Turing Test

The Turing Test was important because it:

1. Provided an **observable criterion** for machine intelligence.
2. Shifted the discussion from philosophical definitions of “thinking” to **machine behaviour**.
3. Influenced research in **natural language processing and conversational AI**.
4. Became one of the most famous ideas in the history of AI.

## Limitations

The Turing Test has also been criticized because:

- Passing a conversation test does not necessarily prove genuine understanding.
- A system might imitate human behaviour without actually reasoning like a human.
- The test focuses mainly on **behaviour and language**.
- It does not directly measure consciousness, emotions, or understanding.

## Chinese Room Argument

The **Chinese Room Argument** was proposed by philosopher **John Searle in 1980**. It was designed as a criticism of the claim that executing an appropriate computer program is sufficient for genuine understanding.

### Basic idea

Imagine a person who does **not understand Chinese** sitting inside a room.

The person has:

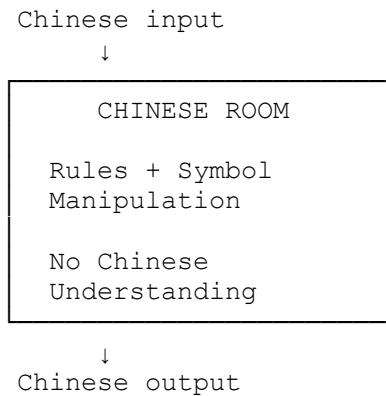
- A set of rules written in English.
- Chinese symbols entering the room.
- Instructions explaining how to manipulate those symbols.
- No knowledge of the Chinese language.

People outside the room send Chinese questions into the room. The person follows the rules and sends back appropriate Chinese answers.

To people outside, it may appear that the person inside **understands Chinese**.

However, the person actually does not understand Chinese—they are simply manipulating symbols according to rules.

## Representation



## Searle's main argument

Searle distinguished between:

- **Syntax** → manipulating symbols according to formal rules.
- **Semantics** → understanding the meaning of those symbols.

His argument is that a computer program operates through **syntax**, but merely executing a program does not necessarily give the computer **semantic understanding**.

Therefore:

**Simulation of understanding is not necessarily the same as actual understanding.**

## Turing Test vs Chinese Room

| Turing Test                                          | Chinese Room                           |
|------------------------------------------------------|----------------------------------------|
| Proposed by <b>Alan Turing</b>                       | Proposed by <b>John Searle</b>         |
| Introduced in <b>1950</b>                            | Introduced in <b>1980</b>              |
| Tests intelligent <b>behavior</b>                    | Questions genuine <b>understanding</b> |
| Based on conversation                                | Based on symbol manipulation           |
| If machine is indistinguishable from a human, it may | Producing appropriate answers does not |

| <b>Turing Test</b>                    | <b>Chinese Room</b>                                  |
|---------------------------------------|------------------------------------------------------|
| be considered intelligent             | necessarily mean understanding                       |
| Supports a behavioural approach to AI | Criticizes strong claims about machine understanding |

The historical development of AI moved from early theoretical ideas about machine intelligence to symbolic reasoning, expert systems, machine learning, deep learning, and modern generative AI. **Turing's 1950 proposal** provided an influential behavioural approach to machine intelligence, while **Searle's Chinese Room argument** raised an important philosophical question: **Can a machine that manipulates symbols actually understand their meaning?**

Together, the **Turing Test** and **Chinese Room Argument** remain fundamental concepts for understanding both the development of AI and the debate over what it means for a machine to be genuinely intelligent.

## Problem Reduction and State Space Search

These are important techniques in **Artificial Intelligence problem solving**, where a complex problem is represented in a form that allows an AI system to find a solution systematically.

### Problem Reduction

**Problem reduction** is a technique of breaking a **complex problem into smaller and simpler sub problems** until the resulting sub problems become easy to solve.

The basic idea is:

**Complex problem → Smaller subproblems → Simple problems → Solution**

### Example

Suppose an AI system has to **travel from City A to City D**.

Instead of solving the entire problem at once:

A → D

it can reduce it into smaller problems:

A → B

B → C

C → D

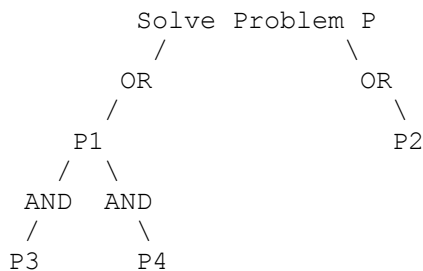
Solving these smaller problems together gives the solution to the original problem.

## AND-OR Representation

Problem reduction is commonly represented using an **AND-OR graph**.

- **OR node:** Any one of the alternatives can solve the problem.
- **AND node:** All subproblems must be solved to solve the original problem.

Example:



If P<sub>1</sub> is selected, both P<sub>3</sub> and P<sub>4</sub> must be solved.

## Advantages

1. Makes complex problems easier to understand.
2. Reduces computational complexity in many cases.
3. Allows individual sub problems to be solved independently.
4. Useful in planning, diagnosis and game playing.
5. Encourages modular problem solving.

## Applications

- Medical diagnosis
- Automated planning
- Game playing
- Theorem proving
- Expert systems
- Scheduling

# State Space Search

**State space search** is a problem-solving method in which a problem is represented as a collection of possible **states**, and the AI searches for a sequence of actions that transforms the **initial state** into the **goal state**.

## Basic components

A state-space problem generally contains:

1. **Initial state** – Starting condition.
2. **Goal state** – Desired final condition.
3. **Operators/Actions** – Operations that change one state into another.
4. **State space** – Collection of all possible states.
5. **Solution path** – Sequence of actions from initial state to goal state.

## Example: 8-Puzzle

Consider an 8-puzzle:

| Initial State |   | Goal State |
|---------------|---|------------|
| 1 2 3         |   | 1 2 3      |
| 4 5 6         | → | 4 5 6      |
| 7 _ 8         |   | 7 8 _      |

The blank space can be moved in different directions.

Each movement produces a **new state**. The AI searches through these states until it reaches the goal.

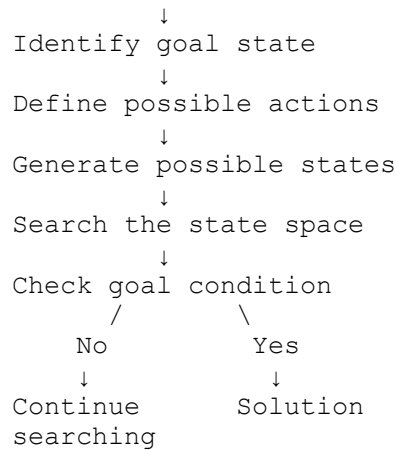
```
Initial
|
+----- State 1
|
+----- State 2
|           |
|           +----- State 4
|
+----- State 3
|
+----- Goal
```

# Problem-Solving Algorithm

A **problem-solving algorithm** is a systematic procedure used by an AI agent to find a sequence of actions that reaches a goal.

## General steps

```
Start
↓
Define the problem
↓
Identify initial state
```



## General algorithm

**Step 1:** Define the problem.

**Step 2:** Specify the initial state.

**Step 3:** Specify the goal state.

**Step 4:** Define the possible operators/actions.

**Step 5:** Generate successor states.

**Step 6:** Search the state space.

**Step 7:** Check whether the current state satisfies the goal.

**Step 8:** If the goal is reached, return the solution path; otherwise continue searching.

## Search Strategies

State-space search can use different strategies.

### Uninformed Search

These methods do not use additional knowledge about how close a state is to the goal.

- **Breadth-First Search (BFS)**
- **Depth-First Search (DFS)**
- Uniform-Cost Search
- Depth-Limited Search
- Iterative Deepening Search

## Informed Search

These methods use additional information, usually a **heuristic**, to guide the search.

- Greedy Best-First Search
- **A\*** Search
- Hill Climbing
- Beam Search

## 5. Problem Reduction vs State Space Search

| Problem Reduction                         | State Space Search                               |
|-------------------------------------------|--------------------------------------------------|
| Breaks a problem into smaller subproblems | Searches through possible states                 |
| Uses problem decomposition                | Uses states and operators                        |
| Often represented using AND-OR graphs     | Often represented using state-space graphs/trees |
| Focuses on reducing complexity            | Focuses on finding a path to the goal            |
| Useful when a problem can be decomposed   | Useful when actions produce different states     |

**Problem reduction simplifies a difficult problem by dividing it into manageable sub problems, while state space search solves a problem by exploring possible states and actions until the goal state is reached. Both are fundamental AI problem-solving techniques and form the basis for algorithms such as BFS, DFS, and A\*.**

## Major Components of AI and Search Process

Artificial Intelligence consists of several major components that enable machines to **perceive, learn, reason, solve problems, and make decisions**. The **search process** is one of the fundamental techniques used by AI to find solutions to problems.

## Major Components of Artificial Intelligence

### Knowledge Representation

Knowledge representation is the process of storing information about the real world in a form that an AI system can understand and use.

It represents:

- Facts
- Rules
- Objects
- Relationships

- Situations

**Examples:** Logic, semantic networks, frames, production rules.

**Example:**

Fact:        Birds have wings.  
Fact:        Sparrow is a bird.  
Conclusion: Sparrow has wings.

## **Reasoning**

Reasoning allows an AI system to **draw conclusions from available knowledge**.

AI can use:

- Deductive reasoning
- Inductive reasoning
- Probabilistic reasoning

**Example:**

All humans are mortal.  
Ravi is a human.  
Therefore, Ravi is mortal.

## **Learning**

Machine learning enables an AI system to **learn from data and experience** instead of being explicitly programmed for every situation.

Major types include:

- Supervised learning
- Unsupervised learning
- Reinforcement learning

**Example:** A spam filter learns to identify spam emails from previously classified emails.

## **Problem Solving**

AI systems solve problems by determining a sequence of actions that leads from an **initial state to a goal state**.

Common techniques include:

- State-space search

- Problem reduction
- Heuristic search
- Means-ends analysis

## 5. Perception

Perception allows AI systems to obtain information from the environment through sensors or input data.

Examples:

- Image recognition
- Speech recognition
- Object detection
- Facial recognition

**Example:** A self-driving vehicle uses cameras and sensors to detect roads, vehicles and pedestrians.

## . Natural Language Processing (NLP)

NLP enables computers to **understand, process and generate human language**.

Applications include:

- Chat bots
- Machine translation
- Voice assistants
- Text summarization
- Sentiment analysis

## Planning and Decision Making

Planning determines **what actions should be performed and in what order** to achieve a particular goal.

**Example:**

A delivery robot may plan:

Warehouse → Road A → Road B → Customer

It selects a suitable route while considering obstacles and costs.

## Robotics

Robotics combines AI with mechanical systems to allow machines to **sense their environment, make decisions and perform physical actions.**

Applications include:

- Industrial robots
- Medical robots
- Autonomous vehicles
- Service robots

## Search Process in AI

The **search process** is a systematic method of exploring possible solutions to a problem.

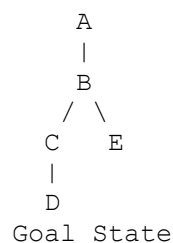
An AI problem is generally represented using:

- **Initial state**
- **Goal state**
- **Operators/actions**
- **State space**
- **Path/solution**

### Example

Suppose a robot needs to move from position A to position D.

Initial State



The AI searches through the available states to find a path:

**A → B → C → D**

## Steps in the AI Search Process

### Step 1: Define the Problem

Clearly specify what needs to be solved.

## Step 2: Identify the Initial State

Determine where the problem begins.

## Step 3: Define the Goal State

Determine the desired final condition.

## Step 4: Define Operators

Specify the actions that can transform one state into another.

## Step 5: Generate Successor States

Apply available operators to the current state to produce new states.

## Step 6: Select a State for Expansion

The search strategy determines which state should be explored next.

## Step 7: Test for the Goal

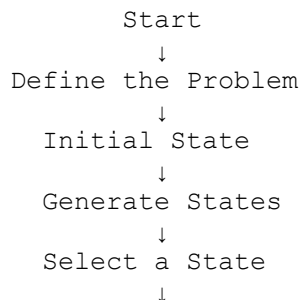
Check whether the selected state satisfies the goal condition.

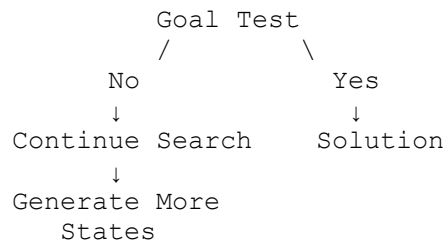
- If **yes**, return the solution.
- If **no**, continue searching.

## Step 8: Return the Solution

The sequence of actions from the initial state to the goal state is the **solution path**.

## General Search Process





# Search Strategies

AI search techniques are broadly divided into two categories.

## A. Uninformed Search

No additional information about the goal is used.

Examples:

- **Breadth-First Search (BFS)**
- **Depth-First Search (DFS)**
- Uniform-Cost Search
- Depth-Limited Search
- Iterative Deepening Search

## B. Informed Search

Uses additional knowledge called a **heuristic** to guide the search toward the goal.

Examples:

- Greedy Best-First Search
- **A\* Search**
- Hill Climbing
- Beam Search

## 6. Important Characteristics of a Search Algorithm

A good AI search algorithm is generally evaluated using:

| Property                | Meaning                                                   |
|-------------------------|-----------------------------------------------------------|
| <b>Completeness</b>     | Whether it is guaranteed to find a solution if one exists |
| <b>Optimality</b>       | Whether it finds the best/least-cost solution             |
| <b>Time Complexity</b>  | Amount of computation required                            |
| <b>Space Complexity</b> | Amount of memory required                                 |

The **major components of AI** include **knowledge representation, reasoning, learning, problem solving, perception, NLP, planning, decision making, and robotics.**

The **search process** provides a systematic way to solve problems by exploring possible states. It starts with an **initial state**, generates possible successor states using operators, performs a **goal test**, and continues until a suitable solution is found.